

WHAT THIS COURSE TEACHES

Every controlled process has a PID loop at its core. Flow control, pressure regulation, temperature setpoints, level management, pH dosing — the algorithm is the same everywhere, and most engineers who configure these loops have never been taught why the parameters do what they do. They tune by trial and error, accept whatever oscillation the process will tolerate, and move on.

This course teaches PID from the mathematical foundation through practical tuning, cascade control, and digital implementation in PLCs. Engineers who complete it can look at a process response and diagnose why a loop is oscillating, sluggish, offset, or windup-limited — without guessing.

CHAPTER SEQUENCE

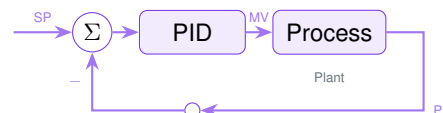
1. **Loop Fundamentals** — Closed-loop control, error signal, feedback vs. feed-forward, why open-loop control fails.
2. **Process Dynamics** — First-order lag, dead time, process gain, integrating processes, the FOPDT model.
3. **PID Action** — Proportional band vs. gain, integral (reset) action and windup, derivative (rate) action and filtering.
4. **Tuning Methods** — Ziegler-Nichols open and closed loop, IMC-based tuning (Lambda), Cohen-Coon, ITAE criteria.
5. **Digital PID Implementation** — Velocity vs. position form, sampling time selection, anti-windup strategies, bumpless transfer.
6. **Cascade Control** — Inner/outer loop architecture, when cascade helps and when it does not, commissioning sequence.
7. **PLC Implementation** — Standard PID function block (IEC 61131-3 PID_FC), manual/auto transfer, mode handling.
8. **Troubleshooting** — Oscillation diagnosis, offset analysis, saturation, valve stiction, sensor lag, process nonlinearity.
9. **Advanced Topics** — Feedforward control, ratio control, dead time compensation (Smith predictor), split-range.
10. **Process Lab** — Step test analysis, FOPDT parameter extraction, Lambda tuning exercise, cascade commissioning.

THE STORY

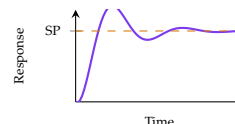
The PID algorithm is 80 years old and remains the dominant control strategy in every industry. Advanced process control (model predictive control, neural networks) exists and is applied in refineries and chemical plants where the economics justify the engineering cost. For the vast majority of industrial processes — water treatment, HVAC, food & beverage, utilities, discrete manufacturing — PID is the right tool and will remain so.

The knowledge gap is in tuning and diagnosis. Many engineers can configure a PID block in their HMI. Almost none have been taught the Laplace transform intuition that explains why integral action causes windup, why derivative action amplifies noise, and why a loop that was tuned on a full tank behaves differently when the tank is empty. This course fills that gap.

Closed-Loop Control



Step Response



Certification Relevance

- **ISA CCST (Level III)** — Process control loops and tuning are core exam topics at the Specialist level.
- **ISA CAP** — Certified Automation Professional: process dynamics and control design are heavily tested.
- **PE (Chemical/Control)** — P.E. exam control systems section covers PID analysis and stability.

Next Steps (Certs)

1. Complete PID + IEC 61131-3 courses as a pair.
2. Review ISA-5.1 (Instrumentation symbols) and ISA-75.01 (control valve sizing).
3. Apply at isa.org for CCST or CAP examination.
4. Five years of experience required for CAP; CCST has no experience prerequisite.

10 Chapters 600+ Questions 4 Methods

